Class 9 CHEMISTRY – Matter in Our Surroundings (NCERT Solutions)

Q1. Convert the following temperatures to Celsius scale

Answer:

Formula:
°C = K − 273

(a) 293 K = 293 − 273 = **20°C**
(b) 470 K = 470 − 273 = **197°C**

Q2. Convert the following temperatures to Kelvin scale

Answer:

Formula:
K = °C + 273

(a) 25°C = 25 + 273 = **298 K**
(b) 373°C = 373 + 273 = **646 K**

Q3. Give reasons

(a) Naphthalene balls disappear because they undergo **sublimation** (solid → gas).

(b) Smell of perfume spreads due to **diffusion of gas particles**.

Q4. Increasing order of forces of attraction

Answer:

Oxygen < Water < Sugar

(Gases < Liquids < Solids)

Q5. Physical state of water

Answer:

(a) 25°C → Liquid
(b) 0°C → Solid + Liquid (both)
(c) 100°C → Liquid + Gas (both)

Q6. Give reasons

(a) Water is liquid because it has definite volume but no definite shape.

(b) Iron almirah is solid because it has definite shape and volume and is rigid.

Q7. Why ice cools more than water at same temperature?

Answer:

Ice absorbs **latent heat of fusion** while melting, thus produces more cooling.

Q8. What causes more burns?

Answer:

Steam causes more burns because it contains **latent heat of vaporisation**.

Q9. Why desert cooler works better on hot dry day?

Answer:

Evaporation is faster in dry air, so cooling effect is more.

Q10. Convert temperatures

(a) 300 K = 27°C
(b) 573 K = 300°C

Q11. Physical state of water

(a) 250°C → Gas
(b) 100°C → Liquid + Gas

Q12. Why temperature remains constant during change of state?

Answer:

Heat supplied is used to change the state (latent heat), not temperature.

Q13. How to liquefy gases?

Answer:

By **increasing pressure** and **decreasing temperature**.

Q14. Why earthen pot becomes cool?

Answer:

Water evaporates through pores → causes cooling.

Q15. Why palm feels cold with acetone/petrol?

Answer:

They evaporate quickly, absorbing heat from palm.

Q16. Why tea cools faster in saucer?

Answer:

Larger surface area → faster evaporation → quicker cooling.

Q17. Clothes in summer

Answer:

Wear **light-coloured, cotton clothes** (absorb sweat and allow evaporation).

Density & States of Matter

Q18. Increasing order of density

Answer:

Air < Exhaust < Cotton < Water < Honey < Chalk < Iron

Q19. Differences between states of matter

Property	Solid	Liquid	Gas
Shape	Fixed	No fixed	No fixed
Volume	Fixed	Fixed	Not fixed
Compressibility	Very low	Low	High
Density	High	Medium	Low

Q20. Reasons

(a) Gas fills vessel → particles move freely
(b) Gas exerts pressure → constant collisions
(c) Table is solid → rigid and fixed shape
(d) Hard to move in solid → strong forces of attraction

Q21. Why ice floats on water?

Answer:

Ice has **lower density** than water due to open structure.

Basic Concepts of Matter

Q22. Which are matter?

Answer:

Matter: Chair, air, almonds, lemon water, smell of perfume

Not matter: Love, hate, thought, cold

Q23. Smell spreads faster in hot food

Answer:

Higher temperature → faster particle movement → faster diffusion

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Q24. Diver cuts through water

Answer:

Shows **weak forces of attraction in liquids**

Q25. Characteristics of particles of matter

Answer:

- Very small
- Have spaces between them
- In constant motion
- Attract each other

Key Points for Revision

- K = °C + 273
- Diffusion increases with temperature
- Evaporation causes cooling
- Solids > Liquids > Gases (force of attraction)
- Density = mass / volume

Class 9 CHEMISTRY – Is Matter Around Us Pure (NCERT Solutions)

Q1. Separation techniques

Answer:

(a) Sodium chloride from water → **Evaporation**
(b) Ammonium chloride + sodium chloride → **Sublimation**
(c) Metal pieces in oil → **Filtration / Magnetic separation**
(d) Pigments from petals → **Chromatography**
(e) Butter from curd → **Centrifugation (churning)**
(f) Oil from water → **Separating funnel**
(g) Tea leaves from tea → **Filtration**
(h) Iron pins from sand → **Magnetic separation**
(i) Wheat grains from husk → **Winnowing**
(j) Mud from water → **Sedimentation and decantation**

Q2. Steps to make tea (using given words)

Answer:

Water is taken as the **solvent**. Tea leaves and sugar are added as **solutes**.
Sugar is **soluble** and **dissolves** in water forming a **solution**.
Tea leaves are **insoluble** and do not dissolve.
The mixture is filtered; the liquid obtained is **filtrate** (tea), and tea leaves left behind are **residue**.

Q4. Definitions

(a) Saturated solution:
A solution in which no more solute can dissolve at a given temperature.

(b) Pure substance:
A substance made of only one type of particles (e.g., gold, oxygen).

(c) Colloid:
A mixture with intermediate particle size that does not settle (e.g., milk).

(d) Suspension:
A mixture where particles are large and settle down (e.g., muddy water).

Q5. Homogeneous / Heterogeneous

Answer:

Homogeneous: Soda water, air, vinegar, filtered tea
Heterogeneous: Wood, soil

Q6. How to confirm pure water?

Answer:

- Pure water has a **fixed boiling point (100°C)** and **freezing point (0°C)**
- It leaves **no residue on evaporation**

Q7. Pure substances

Answer:

Pure substances:
Ice, Iron, Hydrochloric acid, Calcium oxide, Mercury

Not pure:
Milk, Brick, Wood, Air

Q8. Identify solutions

Answer:

Solutions: Sea water, Air, Soda water

Not solutions: Soil, Coal

Q9. Tyndall effect

Answer:

Shows Tyndall effect:
Milk, Starch solution

Does not show:
Salt solution, Copper sulphate solution

Q10. Classification

Answer:

Elements: Sodium, Silver, Tin, Silicon

Compounds: Calcium carbonate, Methane, Carbon dioxide

Mixtures: Soil, Sugar solution, Coal, Air, Soap, Blood

Q11. Chemical changes

Answer:

Chemical changes:
Growth of plant, Rusting, Cooking, Digestion, Burning of candle

Physical changes:
Mixing iron & sand, Freezing of water

Extra Questions

Q12. Chemical vs Physical changes

Answer:

Chemical: Rusting, electrolysis of water, burning

Physical: Cutting trees, melting butter, boiling water, dissolving salt, fruit salad

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Q13. Homogeneous vs Heterogeneous

Answer:

Homogeneous	Heterogeneous
Uniform composition	Non-uniform
Single phase	Multiple phases

Homogeneous	Heterogeneous
Example: salt solution	Example: sand + water

Q14. Sol, Solution, Suspension

Answer:

Property	Solution	Colloid (Sol)	Suspension
Particle size	Small	Medium	Large
Visibility	Not visible	Not visible	Visible
Settling	No	No	Yes

Q15. Concentration of solution

Answer:

Mass of solute = 36 g
Mass of solution = 36 + 100 = 136 g

Concentration = (36 / 136) × 100
= **26.47%**

Q16. What is a substance?

Answer:

A substance is a pure form of matter having definite composition and properties.

Q17. Differences (again for exam clarity)

Homogeneous mixture:

- Uniform
- No visible boundaries

Heterogeneous mixture:

- Non-uniform
- Visible boundaries

Key Points for Revision

- Evaporation, filtration, chromatography → separation
- Pure substance = element or compound
- Solution = homogeneous mixture
- Colloids show Tyndall effect
- Chemical change = new substance formed

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Class 9 CHEMISTRY – Atoms and Molecules (NCERT Solutions)

Q1. Percentage composition of compound

Answer:

Given:
Total mass = 0.24 g
Mass of Boron = 0.096 g
Mass of Oxygen = 0.144 g

Percentage of Boron:
= (0.096 / 0.24) × 100 = **40%**

Percentage of Oxygen:
= (0.144 / 0.24) × 100 = **60%**

Final Answer:
Boron = 40%, Oxygen = 60%

Q2. Mass of CO_2 formed

Answer:

Given:
3 g carbon + 8 g oxygen → 11 g CO_2

If oxygen = 50 g, carbon is limiting reactant

So, CO_2 formed = **11 g**

Law used:
Law of constant proportions

Q3. Polyatomic ions

Answer:

Polyatomic ions are groups of atoms carrying a charge.

Examples:
NH_4^+ (Ammonium), SO_4^{2-} (Sulphate), NO_3^- (Nitrate)

Q4. Chemical formulae

Answer:

(a) Magnesium chloride → **$MgCl_2$**
(b) Calcium oxide → **CaO**
(c) Copper nitrate → **$Cu(NO_3)_2$**
(d) Aluminium chloride → **$AlCl_3$**
(e) Calcium carbonate → **$CaCO_3$**

Q5. Elements present

Answer:

(a) Quick lime → Calcium, Oxygen
(b) Hydrogen bromide → Hydrogen, Bromine
(c) Baking powder → Sodium, Hydrogen, Carbon, Oxygen
(d) Potassium sulphate → Potassium, Sulphur, Oxygen

Q6. Molar mass

Answer:

(a) C_2H_2 = (2×12) + (2×1) = **26 g/mol**
(b) S_8 = 8×32 = **256 g/mol**
(c) P_4 = 4×31 = **124 g/mol**
(d) HCl = 1 + 35.5 = **36.5 g/mol**
(e) HNO_3 = 1 + 14 + (3×16) = **63 g/mol**

Extra Practice

Q7. Molecular masses

Answer:

H_2 = 2 g/mol
O_2 = 32 g/mol
Cl_2 = 71 g/mol
CO_2 = 44 g/mol
CH_4 = 16 g/mol

C_2H_6 = 30 g/mol
C_2H_4 = 28 g/mol
NH_3 = 17 g/mol
CH_3OH = 32 g/mol

Q8. Formula unit masses

Answer:

ZnO = 65 + 16 = **81 u**
Na_2O = (2×23) + 16 = **62 u**
K_2CO_3 = (2×39) + 12 + (3×16) = **138 u**

Q9. Write formulae

Answer:

(i) Sodium oxide → **Na_2O**
(ii) Aluminium chloride → **$AlCl_3$**
(iii) Sodium sulphide → **Na_2S**
(iv) Magnesium hydroxide → **$Mg(OH)_2$**

Q10. Name the compounds

Answer:

(i) $Al_2(SO_4)_3$ → Aluminium sulphate
(ii) $CaCl_2$ → Calcium chloride
(iii) K_2SO_4 → Potassium sulphate
(iv) KNO_3 → Potassium nitrate
(v) $CaCO_3$ → Calcium carbonate

Q11. Chemical formula

Answer:

A chemical formula represents the **types and number of atoms** present in a compound.

Q12. Number of atoms

Answer:

(i) H_2S → 3 atoms
(ii) PO_4 → 5 atoms

Q13. Atomic mass unit

Answer:

1 atomic mass unit (u) is defined as **1/12th the mass of a carbon-12 atom**.

Q14. Why atoms cannot be seen?

Answer:

Atoms are extremely small ($\approx 10^{-10}$ m), so they cannot be seen with naked eyes.

Key Points for Revision

- % composition = (part / total) × 100
- Molar mass = sum of atomic masses
- Chemical formula shows composition
- Atoms are extremely small
- Polyatomic ions carry charge

Class 9 CHEMISTRY – Structure of the Atom (NCERT Solutions)

Q1. Compare electrons, protons and neutrons

Answer:

Particle	Charge	Mass	Location
Electron	−1	Very small (~1/2000 of proton)	Outside nucleus
Proton	+1	1 u	Inside nucleus
Neutron	0	1 u	Inside nucleus

Q2. Limitations of J.J. Thomson's model

Answer:

- Could not explain scattering results
- No clear idea of nucleus
- Could not explain stability of atom

Q3. Limitations of Rutherford's model

Answer:

- Could not explain stability of electrons
- Did not explain arrangement of electrons in orbits

Q4. Bohr's model of atom

Answer:

- Electrons revolve in fixed circular orbits (shells)
- Each orbit has fixed energy
- No energy loss in stable orbits
- Energy emitted/absorbed when electrons jump between shells

Q5. Comparison of models

Answer:

- Thomson → atom as positive sphere with electrons
- Rutherford → nucleus at center, electrons around
- Bohr → fixed orbits with energy levels

Q6. Rules for distribution of electrons

Answer:

- Maximum electrons in shell = $2n^2$
- Outer shell maximum = 8 electrons
- Shells filled step by step (K, L, M…)

Q7. Valency (examples)

Answer:

Valency is combining capacity of an atom.

Silicon (14): configuration 2,8,4 → valency = 4
Oxygen (8): configuration 2,6 → valency = 2

Q8. Definitions

Answer:

(i) Atomic number (Z): number of protons
Example: Oxygen = 8

(ii) Mass number (A): protons + neutrons
Example: Carbon = 12

(iii) Isotopes: same Z, different mass
Example: ^{1}H, ^{2}H, ^{3}H

(iv) Isobars: same mass, different Z
Example: ^{40}Ar, ^{40}Ca

Uses of isotopes:

- Medical treatment (Iodine-131)
- Carbon dating

Q9. Why Na^+ has filled shells?

Answer:

Na (11): 2,8,1
Na^+ loses 1 electron → 2,8 (stable configuration)

Q10. Average atomic mass of bromine

Answer:

= (79 × 49.7 + 81 × 50.3) / 100
= (3926.3 + 4074.3)/100
= **80.0 u**

Q11. Percentage of isotopes

Answer:

Let % of ^{16}X = x
Then % of ^{18}X = (100 − x)

16x + 18(100 − x) = 1620

16x + 1800 − 18x = 1620
−2x = −180
x = 90

Answer:
^{16}X = 90%, ^{18}X = 10%

Q12. If Z = 3

Answer:

Element = Lithium
Configuration = 2,1
Valency = **1**

Q13. Mass numbers

Answer:

X = 6 + 6 = 12
Y = 6 + 8 = 14

Relation: **Isotopes**

Q14. True / False

(a) F
(b) F
(c) T
(d) F

Q15. Discovery

Answer:
(a) Atomic nucleus ✓

Q16. Isotopes

Answer:
(c) Different number of neutrons ✓

Q17. Valence electrons in Cl^-

Answer:
(b) 8 ✓

Q18. Electronic configuration of sodium

Answer:
(d) 2,8,1 ✓

Extra Questions

Q19. H, D, T particles

Answer:

Isotope	Protons	Neutrons	Electrons
H	1	0	1
D	1	1	1
T	1	2	1

Q20. Atomic number & charge

Answer:

Atomic number = 8
Charge = 0 (neutral atom)

Q21. Mass number

Answer:

Oxygen = 16
Sulphur = 32

Q22. Valency

Answer:

Chlorine = 1
Sulphur = 2
Magnesium = 2

Q23. Electron distribution

Answer:

Carbon (6) → 2,4
Sodium (11) → 2,8,1

Q24. Total electrons if K and L full

Answer:

K = 2, L = 8 → Total = **10 electrons**

Q25. Neutral atom (Thomson)

Answer:

Positive and negative charges balance → atom is neutral

Q26. Particle in nucleus (Rutherford)

Answer:

Protons are present in nucleus

Q27. Bohr model (3 shells)

Answer:

Atom has K, L, M shells with electrons in fixed orbits

Q28. Alpha scattering with other metal

Answer:

Similar results will be observed → confirms nucleus concept

Key Points for Revision

- Atomic number = protons
- Mass number = protons + neutrons
- Isotopes → same Z, different mass
- Bohr model explains stability
- Valency = combining capacity
